Multilingual Entity Matching

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Abstract The aim of this paper is to explore methods of multilingual entity matching. Name matching is currently the main technique used for entity resolution. When dealing with entities having features recorded in different languages and with different alphabets the basic approaches have serious limitation. The basic name matching approach using string comparison metrics is enriched with phonetic rules and with relational information. The results show that the approach using transliteration enhanced by phonetic matching provides with the best performance.

1 Introduction

Entity resolution (also known as *record linkage* [23], *reference reconciliation* [21], *object matching*) is the task of finding records from one or multiple databases, referring to the same real-world entity [22]. Entity resolution in a single database case is sometimes called *duplicate detection* or *deduplication* [6].

As the amount of online data increases and duplicate messages are flooded, entity resolution needs more attention. Practically, information seekers distinguish entities by themselves when there is confusion. But it is important to link entities across platforms to provide users with advanced services.

In this paper, we introduce current approaches and examine their validity. Then we discuss limitations of such approaches and introduce a new problem of multilin-

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gual entity resolution. We propose a model to solve the multilingual entity resolution and present the performance.

2 Background

Kouki et al. [12] presented an entity resolution system which constructs familial networks based on healthcare records from patients from one family. Such healthcare records can contain one patient having a 15-year daughter named Lisa suffering from a high blood pressure and contain another patient having a 16-year old daughter named Liza with a type 1 diabetes. Identifying if both 15 years old Liza and 16 years old Lisa are actually the same person, assuming the data (names, ages, etc.) can contain errors, is the task of entity resolution.

Kouki et al. state that using only attribute similarities is problematic because the data can be insufficient and unreliable (patients can incorrectly report names, ages and forget to mention some family members). As a solution to this problem, the authors add *relational information* to the feature space. Relational information and *collective matching* can improve the results of the matching [12, 4, 8].

While there is already some research on collective entity resolution [4, 8], there is still a possibility to improve the performance of matching by using better attribute similarity measures. Improving this way can be favorable when the computational power is limited. In this paper we want to focus on the *name matching* aspect of entity resolution.

Name matching is also part of entity resolution where name strings are linked if they are instances of the same name [18, 17]. Given a name represented by the string A in one language and a name represented by the string B in possibly other language, a name matching algorithm should tell if A and B represent the same person or give a probability of this.

One alternative to exact string matching is to convert name strings to some common phonetic representation of the names and comparing the phonetic representations. Some of the phonetic representations are Soundex [20], Match Rating Approach [16], Daitch-Mokotoff Soundex [15], Beider-Morse Phonetic Matching [3], Double Metaphone [19]. We evaluate the applicability of the latter in this paper.

- "Soundex" algorithm [20] is the ancestor of phonetic name matching algorithms . Soundex maps names to a special code consisting of a letter and three digits. The letter is the first letter of the name and the digits describes approximatively the consonants of the name. The initial aim of Soundex was to be easily manually computed and to be applied to paper documents. Its performance is relatively poor compared to more recent development.
- Match Rating Approach (MRA) [16] employs a very basic but straightforward process that transforms the string by deleting the vowels and double letters , thereby reducing the name to a maximum of six characters by retaining the first and the last three characters. After the two names are encoded the MRA compares the two strings. The approach can work for strings in multiple languages.

- Daitch-Mokotoff Soundex approach [15] consists in an alternation of the original Soundex for German and Slavic. In this algorithm names are given to six digit codes. The first letter is coded too (contrast to original Soundex where the first letter was retained as it was). Names can have multiple codes which is different from the original Soundex where names are mapped to only one code.
- Beider-Morse Phonetic Matching [3] is a modification of Daitch-Mokotoff Soundex aimed to work with multiple languages. One of the differences is that the first step is to identify the language, then language specific rules are applied.

After the names were converted to their phonetic representations or if no conversion was done and the names remained as they were given, the two strings should be compared.

- Levenshtein's distance [13] between two strings is a metric representing the number of one character changes (substitute a character by another character, remove a character, insert a character) needed to change one word into another.
- Guth's algorithm was specifically designed to compare names. It takes two names as an input and as an output provides with the probability of the two names are variants of spelling of one name. Guth's algorithm compares two strings character by character, sometimes skipping or backtracking one or two characters.

Name matching techniques are essential for joining data from different sources. Using exact string matching is not enough because one person can be referred to in multiple ways: "Bill Clinton", "William Jefferson Clinton" [5], or even the Russian version "Билл Клинтон" for the multilingual name matching. Matching these possible forms of the name is the task of name matching.

The benefits of using name matching techniques instead of simple string similarity measures (like Levenshtein [13], Jaro [7], Jaro-Winkler [24], etc.) can be different for different datasets. However, once again, using name similarities only can be insufficient to separate two different real-world people having the same name and relational features can help with this.

Applying name matching techniques has some difficulties. Usually, name similarity functions are designed to measure similarity between two words (two first names, two last names, etc.) and not between full names, thus names should be separated into parts and the corresponding words should be found. Without knowing the context (language, naming customs of the person, etc.) it is a hard problem on its own.

Entity resolution has been studied for a long time. Entity resolution methods can be categorized by the level of using relational information in matching, as described in [4]:

- Attribute-only Entity Resolution. Record similarity depends only on the similarities of the attributes.
- Naive Relational Entity Resolution. Record similarity depends on the similarities of the attributes of the two records (as in the previous case) and the similarities of the attributes of the records related to the two records being matched.

- Collective Relational Entity Resolution. Record similarity depends on the similarities of the attributes of the two records (as in the attribute-only case) and the similarity of the records related to the two records being matched.

All of the categories from above can use different *similarity functions* depending on the domain of the data and the structure of the data. For example, Adafre and Rijke [1] use sets of incoming links to measure the similarity of Wikipedia articles, Han et al. [10] match in-text references and use the similarity of texts near the references as one of the similarity criteria, Kouki et al. [12] use name similarities of people names to match references in patient reports. Additionally, *collective* matching criteria can be used [4]: matching of two references can depend on decision about other references.

Often, entity resolution is used to match people or organization references in which case *name matching* techniques can be used.

3 Model Description

The proposed model is based on Probabilistic Soft Logic (PSL) [2] framework. PSL uses *soft truth values* $\in [0,1]$ and *relaxation rules* to encode logical models. The relaxation rules are [11]:

$$\begin{aligned} p \wedge q &= \max(0, p+q-1) \\ p \vee q &= \min(1, p+q) \\ \neg p &= 1-p \end{aligned}$$

As a result of training, the model will be able to give probability of two mentions a and b referring to the same real-world entity: $\mathrm{Same}(a,b)$.

The proposed model consists of a number of PSL rules. All of the rules used can be separated into these types: attribute similarity rules, relational-attribute rules, collective rules, domain rules, prior rules.

3.1 Attribute Similarity Rules

Attribute similarity rules state that if some attribute is similar in two references, then the references should be matched, and if some attribute is not similar in two references, then the references should not be matched. Some examples:

$$
\begin{aligned}
\text{SimName}_{JW}(a,b) &\implies \text{Same}(a,b) \\
\neg\,\text{SimName}_{JW}(a,b) &\implies \neg\,\text{Same}(a,b)
\end{aligned}
$$

$$
\begin{aligned}
\text{SimAge}(a,b) &\implies \text{Same}(a,b) \\
\neg\,\text{SimAge}(a,b) &\implies \neg\,\text{Same}(a,b)
\end{aligned}
$$

Where SimName_{JW} is the Jaro-Winkler name similarity of the two records, SimAge is defined as maximum of the ages divided by the minimum of the two ages.

Kouki et al. note that some attributes like gender should not be an evidence for matching, but can be an evidence against matching:

$$
\begin{aligned}
\text{SameGender}(a,b) &\not\Longrightarrow \text{Same}(a,b) \\
\neg\,\text{SameGender}(a,b) &\implies \neg\,\text{Same}(a,b)
\end{aligned}
$$

3.2 Relational-Attribute Rules

Kouki et al. combine relational features (e.g. has a mother) with attribute features, resulting in rules like:

$$
\begin{aligned}
\text{SimMother}(a,b) &\implies \text{Same}(a,b) \\
\text{SimSisters}(a,b) &\implies \text{Same}(a,b)
\end{aligned}
$$

Where SimMother is defined as the maximum of Levenshtein and Jaro-Winkler similarities of the first names. SimSisters and other multi-valued relationships are defined differently: for a relationship type t (*is a sister of* for SimSisters) and two mentions a, b we find two sets $A = \{x : t(a,x)\}$ and $B = \{y : t(b,y)\}$ which are the sets of mentions related to a and b respectively with the relationship t (for SimSisters A and B will be the sets of all sisters of a and b respectively). The similarity between sets A and B will be the relational-attribute similarity of a and b and is defined as:

$$
\text{Sim}(A,B) = \frac{1}{|A|} \sum_{x \in A} \max_{y \in B} \text{SimName}(x,y)
$$

Assuming relationship is symmetric it is assumed, without loss of generality, that $|A| \leq |B|$.

3.3 Collective Rules

These rules enable the collective matching of references and entities. Collective rules for one-to-one relationships look like this:

$$\mathrm{Rel}(\texttt{Father}, x, a) \wedge \mathrm{Rel}(\texttt{Father}, y, b) \wedge \mathrm{Same}(x, y) \implies \mathrm{Same}(a, b)$$

One-to-many relationships (e.g. *is a sister of*) additionally need to distinguish different relationship participants (e.g. sisters) to avoid matching all participants as one person. Name similarity was chosen as a such separator:

$$\mathrm{Rel}(\texttt{Sister}, x, a) \wedge \mathrm{Rel}(\texttt{Sister}, y, b) \wedge \mathrm{Same}(x, y) \\ \wedge \mathrm{SimName}(a, b) \implies \mathrm{Same}(a, b)$$

3.4 Domain Rules

Some rules follow from the problem setting, a mention from one report can match at most one mention from some other report:

$$\mathrm{FromReport}(a, R_1) \wedge \mathrm{FromReport}(b, R_2) \wedge \mathrm{FromReport}(c, R_2) \\ \wedge \mathrm{Same}(a, b) \implies \neg \mathrm{Same}(a, c)$$

Also, there is a transitivity rule for matching references in different reports:

$$\mathrm{FromReport}(a, R_1) \wedge \mathrm{FromReport}(b, R_2) \wedge \mathrm{FromReport}(c, R_3) \\ \wedge \mathrm{Same}(a, b) \wedge \mathrm{Same}(b, c) \implies \mathrm{Same}(a, c)$$

3.5 Prior Rules

Initially we assume that most of the references do not match which is captured by this rule:

$$\neg \mathrm{Same}(a, b)$$

4 Computing Similarities of English and Russian Names

In this section, we specifically discuss multilingual entity resolution problem with Russian and English names.

4.1 Transliteration

Russian and English languages use different alphabets (Cyrillic and Latin respectively) which makes string similarity functions (like Jaro-Winkler, Levenshtein) unsuitable for comparing names from these languages. To overcome this issue, we convert Russian names to Latin script.

In general, for a name there can be several possible valid equivalents in other language. Such conversion can be done via *transliteration*, *transcription* or *translation*. Transliteration is a more systematic and reversible procedure (Ельцин to Elcin), transcription is a more phonetic conversion focused on preserving the pronunciation of the name (Ельцин to Yeltsin). translation is done via mapping a name from the first language to a traditional equivalent name of the second language (Наталья to Nathaly), if there is one. All of these conversions have different rules and customs for different language pairs [14], see Figure 1.

Different domains can contain names produced by different conversion methods: Russian international passports have names converted from Russian to Latin script using a strict set of transliteration rules which are sometimes different from the transcription rules used in less formal contexts.

Even though transcription can be used more widely, we will use the current Russian transliteration rules for the international passports because they are well defined.

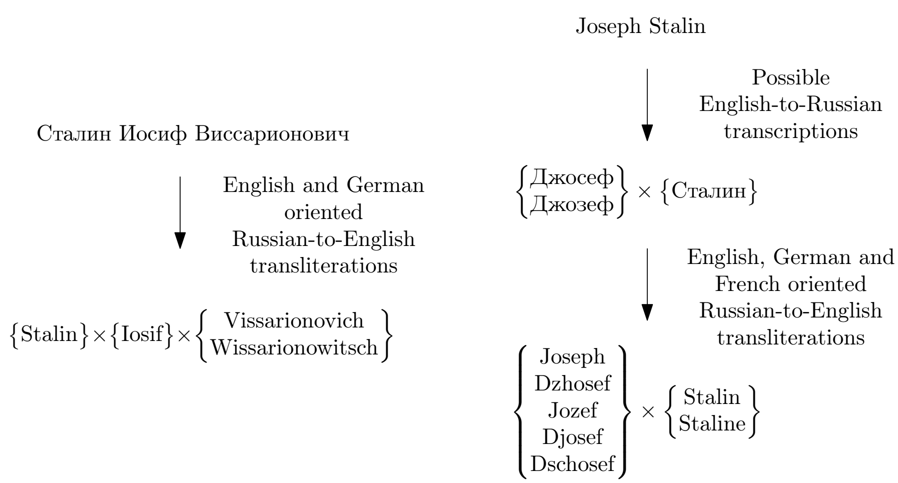

Fig. 1 Cyrillic-Latin and Latin-Cyrillic name variations

4.2 Aligning Names

Names in different languages and cultures can have different structures: Russian names (e.g. Vladimir Vladimirovich Putin) have a given name (Vladimir), patronymic (Vladimirovich) and a family name (Putin), Spanish names (e.g. Pedro Sánchez Pérez-Castejón) have a given name (Pedro) and two family names(Sánchez and Pérez-Castejón), English names (e.g. Theresa Mary May) consist of the first name (Theresa), the last name (May) and optional middle names (Mary). Name variation can be entailed also by deviations of the same name in different languages.

Moreover, a name can appear in different forms: name parts can be omitted (Vladimir Vladimirovich Putin, Vladimir Putin), name parts can be reduced to initials (V. Putin), grammatical transformations like inflection (Vladimira Putina) and so on.

This variety of forms makes extracting name parts and computing name similarities a hard task. To partially overcome this issue, we introduce the *alignment* procedure as a part of computing name similarity.

Full-name similarity of two names with alignment is defined as the maximum similarity of all part permutations (V. Putin, Putin V.) with the second name. This way, similarity between Vladimir Putin and Putina Lyudmila will be high, but not maximum. It is not needed to compute permutations of the second name, assuming the similarity function is symmetric.

4.3 Phonetic Transformations

Converting names between languages can make names less recognizable, especially so when translations or historical transliteration customs are used. For example, historically, letter H (sound /h/) is transcribed from German to Russian as Г (sound /g/). Consequently, converting Hitler to Russian and then to English might result in Gitler. This is not the only example of such inconsistencies. Another example is the Persian name Rostam which is known as Rüstem in Turkish, Röstäm in Tatar, and Рустем in Kazakh languages. Converting to English by dropping diacritics [14] will result in several different names: Rostam, Rustem, Rustam.

Sometimes names have equivalents for names in other languages. For example, Russian name Михаил (Mikhail) has a Ukrainian counterpart Михайло (Mykhailo). Both of these names would be printed in the passport in Soviet Union and the official transliteration would be Mikhail, but nowadays the name is spelled as Mykhailo [25].

In this paper we use Double Metaphone [19] to encode names. Double Metaphone was designed to account for the differences in writings of names in different languages. For example, Double Metaphone codes of both Michael and Maykl (from Майкл) will be equal to MKL.

5 Model Discussion and Improvements Motivation

We can see that SimName predicate is used in many rules, thus the performance of matching might depend heavily on the performance of the name similarity measuring methods. There main obstacles for applying this matching method to other datasets.

The first problem is that SimName is defined as a similarity of the *first* names and SimName_{JW} computes similarities between first names, middle names, last names. Both of these functions are left undefined for the cases where names are represented as one string and it is unknown how to separate names into parts (first name, last name, etc.).

The second problem is that SimName and SimName_{JW} use Jaro-Winkler and Levenshtein distances which will give maximum distances for strings written in two different alphabets (e.g. Latin and Cyrillic). This also limits the applicability of the approach in multi-lingual context.

6 Experiments

The original paper [12] describes the performance of the approach on two datasets, a dataset from the National Institutes of Health (NIH) [9] and a dataset crawled from Wikidata[1]. We generate a similar dataset from Wikidata with people from different countries to focus on multi-lingual matching.

Matching performances are measured with the feature sets used in the original paper for Wikidata: *Names*, *Names + Personal Info*, *Names + Personal Info + Relational Info (1st degree)*. As in the original paper, Gaussian noise generated from a Gaussian distribution ($N(0,0.16)$) is added to features. Full names (e.g. first name and last name in one string) are passed to name similarity functions.

Names feature set: only name similarities with SimName_{JW} are used for matching.

Names + Personal Info feature set: same as Names but with age similarity and gender features.

Names + Personal Info + Relational Info (1st degree) feature set: same as Names + Personal Info but with mother, father, daughters, sons, sisters, brothers, spouses similarity features.

Each feature set was evaluated using two name sets: *Baseline* and *Translit*. In *Baseline* matching, name similarity functions were computing similarities between English versions of the names and $N(0,0.16)$ noise was added as in the original paper. In *Translit* matching, name similarities were computing similarities between English and Russian versions of the names as described in Section 4.1. *Translit* matching was refined by by the alignment procedure described in Section 4.2 and the results are reported as *Translit Align*. *Translit Align* was further enhanced by

[1] https://www.wikidata.org/

phonetic matching techniques described in Section 4.3 and the results are reported as *Translit Align Phonetic*.

Threshold selection technique from the original paper was also implemented. The dataset is split not only to *train* and *test* parts, but a third subset is introduced to find a matching threshold which maximizes the F1 score of the matching.

The performance of the model on different data with different name matching techniques is shown in Table 1. Different and Same classes represent non-matches and matches of the references. The results show that the approach using transliteration enhanced by phonetic matching techniques provides with best results. When the transliteration is corroborated with personal and relational information the performance of the algorithm decreases.

The crawled dataset consists of 470 people from 64 families. Each person is tested for matching against each person. The evaluation (test) dataset split contains 134 people, so there are 134 matching pairs and 62846 non-matching pairs, probability of a match is .002.

Table 1 PSL Model Matching Performance

Feature Set	Total F1	Class	Precision	Recall	F1
Baseline Names	.33	Different	1	1	1
		Same	.25	.50	.33
Translit Names	.61	Different	1	1	1
		Same	.67	.56	.61
Translit Names Align	.60	Different	1	1	1
		Same	.52	.70	.60
Translit Names Align Phonetic	.79	Different	1	1	1
		Same	.83	.75	.79
Baseline Names + PI	.54	Different	1	1	1
		Same	.58	.51	.54
Translit Names + PI	.78	Different	1	1	1
		Same	.84	.60	.70
Translit Names Align + PI	.72	Different	1	1	1
		Same	.71	.74	.73
Translit Names Align Phonetic + PI	.80	Different	1	1	1
		Same	.74	.88	.80
Baseline Names + PI + R1	.62	Different	1	1	1
		Same	.61	.63	.62
Translit Names + PI + R1	.72	Different	1	1	1
		Same	.68	.79	.73
Translit Names Align + PI + R1	.72	Different	1	1	1
		Same	.63	.84	.72
Translit Names Align Phonetic + PI + R1	.71	Different	1	1	1
		Same	.57	.95	.72

6.1 Discussion

Due to the big imbalance in the classes, both precision and recall for the *Different* class (non-matches) are close to 100%. Similar figures were acquired by Kouki et al. Several insights can be derived from these results.

Transliteration from Russian makes less "noise" than $N(0, 0.16)$. *Baseline Names* and *Translit Names* experiments show that the model doing pure name matching of names with $N(0, 0.16)$ noise was less efficient both precision- and recall-wise than the same model working with matching English names and transliterated Russian names.

Aligning improves recall. Aligning procedure described in Section 4.2 improves recall of matching by reordering words in names to increase the similarity. However, aligning can increase the similarity of actually non-matching names decreasing the precision of matching.

Phonetic transformations improve recall. Using Double Metaphone codes for computing similarities (described in Section 4.3) improved recall of matching by bucketing similar names into one phonetic code. However, if there are two different names in the dataset having the same phonetic code, they will have the maximum similarity which can decrease the precision if the two names are eventually considered to be matching.

7 Conclusion

This paper explores methods of multilingual entity matching. The basic name matching approach using string comparison metrics is enriched with phonetics rules and with relational information. The results show how different techniques affect precision and recall. Nevertheless, the presented approaches do not address the issues of name variability in the transliteration/transcription/translation process between languages with different alphabets. This aspect will be explored in a further research.

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